

A LIMIT-POINT CRITERION FOR CANONICAL SYSTEMS AND OBSTACLES TO A DE BRUIJN–NEWMAN REALIZATION

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ABSTRACT. We record, with a short proof for reference, the standard trace-divergence criterion (proposition 4.1) for the limit-point classification of the right endpoint of a two-dimensional canonical system, under Remling’s standing assumptions; it is a direct specialisation of Remling’s Theorem 3.5 and is not a new result. We then audit the “Core Lemma” of the speculative programme [12]. Two findings are negative. First, the naive realization via the logarithmic-derivative Herglotz function $m_t = -H'_t/H_t$ exists but is too weak to be useful: it produces a trace-normalised half-line system that is automatically limit-point, making the endpoint conclusion tautological (section 2). Second, and more importantly, the programme’s proposed equivalence “RH $\iff$ $\text{spec } T_\Phi = \{\gamma_n\}$ ” is *not justified as written*: a self-adjoint operator has real spectrum, so its eigenvalues $\{\gamma_n\}$ (the ordinates of the zeta zeros) record nothing about the *real parts* of those zeros, and the roadmap supplies no theorem recovering that missing information (section 3). Since T_Φ has not been rigorously defined, we do not claim to have disproved the implication for that specific relation; we show only that the stated justification is insufficient. A correct sufficient replacement is a one-directional criterion using the characteristic function $\Xi(z) = \xi(\frac{1}{2} + iz)$; it is at least as hard as RH, imposing requirements not known to follow from RH. The paper yields *no* nontrivial or arithmetically meaningful theorem about the de Bruijn–Newman flow.

Scope note. This paper is a reference note plus a logical-gap and proof-obligation audit of a speculative programme. It does *not* imply the Riemann Hypothesis or any consequence thereof. The only proved statements are the criterion proposition 4.1 and the trivial realization of section 2; the programme’s Core Lemma is shown to be, with only the justification it gives, insufficient for RH, and the corrected problem Open Problem 3.2 is marked **OPEN**.

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1. INTRODUCTION

We work throughout in the Rodgers–Tao convention [6]. Let

$$\Phi_{\text{RT}}(u) = \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \quad u \in \mathbb{R},$$

and define the heat-flow deformation, for $t \in \mathbb{R}$, by

$$(1) \quad H_t(z) = \int_0^\infty e^{tu^2} \Phi_{\text{RT}}(u) \cos(zu) du, \quad \partial_t H_t(z) = -\partial_z^2 H_t(z),$$

so that $H_0(z) = \frac{1}{8} \xi(\frac{1}{2} + \frac{iz}{2})$. Note the *backward* heat equation carries a minus sign; it is the kernel weight e^{tu^2} (not a Gaussian multiplier in the transform variable) that genuinely moves the non-real zeros of H_t toward the real axis as t increases. In particular the naive multiplier $e^{-tz^2} H_0(z)$ would leave the zero set of H_0 unchanged and does *not* realise the de Bruijn–Newman flow. The de Bruijn–Newman constant $\Lambda \in [0, 0.2]$ (lower bound $\Lambda \geq 0$ by Rodgers–Tao [6]; upper bound $\Lambda \leq 0.2$ by Saouter–Gourdon–Demichel [9], using zero-free region data from Platt–Trudgian [10]) is defined by: for $t > \Lambda$, H_t has only real zeros (Laguerre–Pólya class; cf. Newman [7]); for $t < \Lambda$, H_t has non-real zeros. The Riemann Hypothesis is equivalent to $\Lambda = 0$.

Remark 1.1 (Relation to the earlier scaling). A previous draft used the Titchmarsh-scaled kernel $\Phi_{\text{doc}}(x) = 2\Phi_{\text{RT}}(x/2)$ together with a Gaussian factor $e^{-\lambda t^2} \Xi_0(t)$. That factor is nowhere zero and hence leaves the zero set of Ξ_0 unchanged, so it does not implement the flow. Under the correct kernel identity one has $\int_0^\infty e^{\lambda x^2} \Phi_{\text{doc}}(x) \cos(tx) dx = 4H_{4\lambda}(2t)$, whence $\Xi_0(t) = \frac{1}{2} \xi(\frac{1}{2} + it)$ and the critical parameter rescales to $\Lambda/4$. To avoid these pitfalls we use only the Rodgers–Tao normalisation above.

De Branges’ theory [1] associates to a 2×2 positive-semidefinite locally-integrable Hamiltonian $H(x)$ the canonical system

$$(2) \quad J y'(x) = z H(x) y(x), \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

At each regular truncation x the transfer matrix yields a de Branges structure function E_x and space $\mathcal{H}(E_x)$, whose reproducing kernel is an *internal* kernel of that space. After a self-adjoint realization is fixed, the associated spectral transform maps the operator (or relation) part unitarily onto an $L^2(\mathbb{R}, \rho)$ space for a spectral measure ρ ; the Krein–de Branges correspondence [2, 3] relates H , the chain $\{E_x\}$, and ρ . We stress that ρ is *not* the reproducing kernel: the two are distinct objects.

The key structural question for the programme of [12] is: *does a canonical system attached to the de Bruijn–Newman flow satisfy the limit-point condition at $x = +\infty$?* A limit-point right endpoint removes the boundary freedom at that endpoint; it does *not* by itself make the associated relation an essentially self-adjoint operator, since the regular left endpoint $x = 0$ still carries one boundary condition (see proposition 4.1 and corollary 5.3).

What this paper does and does not do. The D1 step of [12] gave a *qualitative* argument. The present paper isolates what is actually provable, and is best read as a research audit rather than a source of new theorems.

- (1) **Reference criterion (section 4).** We restate, with a short proof, the standard trace-divergence limit-point criterion proposition 4.1. This is a direct specialisation of [3, Theorem 3.5]; it is not new and its proof depends on [3].

- (2) **Trivial realization (section 2).** For $t > \Lambda$, the logarithmic derivative $m_t = -H'_t/H_t$ is a generalised Herglotz function, so Remling's inverse theorem [3, Theorem 5.1] produces a *unique trace-normalised* half-line canonical system $\widehat{H}_t$. Being trace-normalised, it has $\int_0^\infty \text{Tr } \widehat{H}_t = +\infty$ automatically and is therefore limit-point by proposition 4.1 — a tautology that carries no arithmetic content.
- (3) **Audit of the Core Lemma (section 3).** According to [12] (its “Core Lemma L”), D1's acceptance criterion is a two-part equivalence: T_Φ limit-point at $+\infty$ *and* a fixed self-adjoint realization having spectrum $\{\gamma_n\}$. We find its stated justification *insufficient as written*: the second conjunct records only the ordinates γ_n , which are real regardless of RH, so self-adjointness plus equality of ordinate sets does *not* by itself imply RH; the roadmap gives no theorem recovering the real-part information (remark 3.1). We do *not* disprove the implication for the (still undefined) T_Φ . A correct sufficient replacement uses the characteristic function $\Xi(z) = \xi(\frac{1}{2} + iz)$: a full zero-divisor identification with Ξ *does* imply RH, as a one-way sufficiency at least as hard as RH (Open Problem 3.2 and remark 3.3). We also note that T_Φ and its Hamiltonian are not yet rigorously defined in [12]; constructing them is the first, unstated part of the problem (remark 3.5). No terminal entire function at $x = \infty$ is assumed (remark 3.7).

Consequently the paper contains *no* arithmetically meaningful limit-point theorem about the de Bruijn–Newman flow, and nothing that bears on RH.

Remark 1.2 (Trace under reparametrisation). Under the admissible reparametrisation $X(x) = \int_0^x \text{Tr } H(s) ds$ with $\tilde{H}(X) = H(x)/\text{Tr } H(x)$ one has $\int \text{Tr } \tilde{H}(X) dX = \int \text{Tr } H(x) dx$. Thus finiteness or infiniteness of the total trace *cannot* be altered by renormalising; in particular one cannot escape the tautology of item (2) by “choosing a different normalisation”. The only genuine freedom is in specifying a *different Hamiltonian* — the putative normalised system H_Φ appearing in the proposed Core Lemma eq. (5), with its own Weyl function — rather than reparametrising a fixed one.

2. A TRIVIAL LOG-DERIVATIVE REALIZATION

We first record that a canonical realization “attached to H_t ” exists by an entirely standard construction — and precisely because it is so cheap, it does not advance the programme.

Proposition 2.1 (Log-derivative Herglotz function). *Fix $t > \Lambda$. Then H_t is a real entire function in the Laguerre–Pólya class, with Hadamard factorisation*

$$(3) \quad H_t(z) = C_t z^{2m} e^{-a_t z^2} \prod_n \left(1 - \frac{z^2}{x_{n,t}^2}\right)^{m_n}, \quad a_t \geq 0, \quad \sum_n \frac{m_n}{x_{n,t}^2} < \infty,$$

where $x_{n,t} > 0$ enumerate the positive nonzero zeros of H_t with integer multiplicities $m_n \in \mathbb{Z}_{\geq 1}$, and $m \in \mathbb{Z}_{\geq 0}$ with $2m$ the order of the zero at the origin. Consequently the logarithmic derivative

$$(4) \quad m_t(z) := -\frac{H'_t(z)}{H_t(z)} = 2a_t z - \frac{2m}{z} + \sum_n m_n \left(\frac{1}{x_{n,t} - z} + \frac{1}{-x_{n,t} - z} \right),$$

the series converging locally uniformly on $\mathbb{C} \setminus Z(H_t)$, is a Herglotz function: $\text{Im } m_t(z) > 0$ for $\text{Im } z > 0$.

Proof. The factorisation eq. (3) is the standard form for the Laguerre–Pólya class (a locally uniform limit of real polynomials with only real roots; see [7] and, for the specific

heat-flow zeros, [8, 6]). Differentiating $\log H_t$ term by term (justified by the local uniform convergence from $\sum_n m_n/x_{n,t}^2 < \infty$) gives eq. (4), each zero pair $\pm x_{n,t}$ contributing exactly once. For $\text{Im } z > 0$ each term $1/(x-z)$ with $x \in \mathbb{R}$ has $\text{Im}(1/(x-z)) = \text{Im } z/|x-z|^2 > 0$, the term $-2m/z$ contributes $2m \text{Im } z/|z|^2 \geq 0$, and $2a_t z$ contributes $2a_t \text{Im } z \geq 0$. Since H_t is nonconstant with at least one nonzero real zero, at least one summand contributes strictly, so $\text{Im } m_t(z) > 0$ (not merely ≥ 0). $\square$

Corollary 2.2 (Automatic — and vacuous — limit-point). *Applying Remling’s inverse spectral theorem [3, Theorem 5.1] to m_t of eq. (4) yields a unique trace-normalised half-line canonical system $\widehat{H}_t$ with $\text{Tr } \widehat{H}_t \equiv 1$ a.e. Because m_t is a non-constant Herglotz function (indeed $\text{Im } m_t > 0$ strictly, and the representing measure of m_t is supported on infinitely many points), the associated system is not a single indivisible interval, so the standing assumptions of proposition 4.1 hold. Since $\text{Tr } \widehat{H}_t \equiv 1$, $\int_0^\infty \text{Tr } \widehat{H}_t dx = +\infty$, and proposition 4.1 places $x = +\infty$ in the limit-point case.*

Separately — and this is a distinct observation — setting $A_t = H_t$, $B_t = -H'_t$ gives $B_t/A_t = m_t$, so $E_t = A_t - iB_t = H_t + iH'_t$ is a Hermite-Biehler function. Indeed, since H_t is real entire, $E_t^\# := \overline{H_t(\bar{z})} - i\overline{H'_t(\bar{z})} = H_t(z) - iH'_t(z)$, so $|E_t(z)|^2 - |E_t^\#(z)|^2 = 4|H_t(z)|^2 \text{Im } m_t(z) > 0$ for $\text{Im } z > 0$ by proposition 2.1. Zero-freeness of E_t in the upper half-plane follows: if $E_t(z_0) = 0$ and $H_t(z_0) \neq 0$ then $\text{Im } m_t(z_0) = -1 < 0$, contradicting the Herglotz property; if $H_t(z_0) = 0$ then $\text{Im } z_0 \leq 0$ since all zeros of H_t are real for $t > \Lambda$. This does not identify E_t as a terminal E-function of the reconstructed half-line system $\widehat{H}_t$; it is only a companion observation about H_t .

Remark 2.3 (Why this does not help). The limit-point conclusion of corollary 2.2 is a tautological consequence of trace normalisation, not a fact about the geometry of a “natural” Hamiltonian: by remark 1.2 every trace-normalised half-line system has infinite total trace, so *every* such realization is limit-point regardless of H_t . The construction uses only that H_t has real zeros with the summability eq. (3); it sees nothing arithmetic and carries no information toward RH. In particular it does *not* exhibit H_t as the terminal A-function of a prescribed canonical chain.

3. AUDIT OF THE PROGRAMME’S CORE LEMMA

The trivial construction of section 2 shows that producing *some* canonical system with H_t in view is automatic. The programme [12] instead proposes a specific spectral criterion, its “Core Lemma L”, which we now audit. We find its stated justification *insufficient as written*, and we record a corrected — necessarily weaker, one-directional — replacement.

3.1. The proposed criterion, and why its justification is insufficient. The roadmap [12] states its D1 acceptance criterion as a two-part equivalence. In our notation:

$$(5) \quad \text{RH} \overset{?}{\iff} (T_\Phi \text{ is limit-point at } +\infty) \\ \text{and } (\text{a fixed self-adjoint realization has spec} = \{\gamma_n\}),$$

where T_Φ is a symmetric relation on a half-line canonical system built (in [12]) from the fixed positive kernel Φ through the cosine-transform relation F1, and $\{\gamma_n\}$ are the imaginary parts of the nontrivial zeros of ζ . We write $\overset{?}{\iff}$, not $\iff$, deliberately.

Remark 3.1 (The proposed implication is not justified as written). A self-adjoint relation has *real* spectrum. If some nontrivial zero were off the critical line, $\rho_0 = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$, its ordinate γ would still be a real number. A statement about the real spectrum $\{\gamma_n\}$

therefore records the ordinates only and is *blind to the real parts* β . Hence “ $\text{spec} = \{\gamma_n\}$ ”, *by itself*, does not imply $\beta = \frac{1}{2}$ and does not imply RH.

We are careful about the scope of this observation. It shows that self-adjointness together with equality of ordinate sets is *insufficient* to force RH, and hence that the roadmap’s stated justification is incomplete. It does *not* disprove the right-to-left implication for the particular relation T_Φ : since T_Φ has not been rigorously defined (remark 3.5), it remains logically possible that a specific construction from the full Φ/ξ data would encode the real parts indirectly, e.g. via a theorem $\text{spec } T_\Phi = \{\gamma_n\} \Rightarrow \text{div } D_{\theta_0} = \text{div } \Xi$. What is missing is exactly such a theorem: an additional argument recovering the real-part information, such as the divisor identity eq. (8). Accordingly eq. (5) as written is not a justified restatement of RH, and the Core Lemma needs this gap filled before it can be attacked.

3.2. The corrected statement: a one-way sufficiency. The object that *does* retain the real parts is the entire function

$$(6) \quad \Xi(z) = \xi\left(\frac{1}{2} + iz\right).$$

A nontrivial zero $\rho = \beta + i\gamma$ of ζ corresponds to the zero $z = \gamma + i(\frac{1}{2} - \beta)$ of Ξ , which is *real* iff $\beta = \frac{1}{2}$. Here div denotes the zero divisor, i.e. the formal sum of zeros counted with their orders. Thus real-rootedness of Ξ is equivalent to RH, and a spectral model implies RH only if it pins down the *full zero divisor* of Ξ , not merely a set of real ordinates.

Open Problem 3.2 (Repair obligations for D1). To turn the programme’s Core Lemma into a well-posed, one-directional sufficiency for RH, the following four obligations must be discharged; each has an explicit PASS condition.

- (1) **Construction (non-circular).** Specify a mathematically checkable map $\Phi \mapsto H_\Phi$, including the exact F1 identity and the normalisation, yielding a locally integrable, positive-semidefinite $H_\Phi: [0, \infty) \rightarrow M_2(\mathbb{R})$ and the minimal symmetric relation T_Φ it induces via eq. (2). *PASS condition.* The datum F1 and the rule $\Phi \mapsto H_\Phi$ must be fixed *independently* of the zero divisor of Ξ , of the ordinates $\{\gamma_n\}$, and of RH: neither the desired spectrum nor any measure manufactured from it may enter as input. Existence *and uniqueness* of the resulting normalised H_Φ must be proved — if one Φ could yield different H_Φ under different admissible choices, “the specific operator T_Φ ” would still be undefined.
- (2) **Standing assumptions and length.** Prove that H_Φ meets the standing assumptions of proposition 4.1 and, *independently*, that $\int_0^\infty \text{Tr } H_\Phi(x) dx = +\infty$ (whence limit-point, by proposition 4.1).
- (3) **Discreteness and characteristic function (constructed, not named).** After fixing $\theta_0 \in [0, \pi)$, prove that the resulting self-adjoint realization has *purely discrete* spectrum — with no absolutely continuous or singular continuous part and no finite real accumulation point. Then construct D_{θ_0} by a *prescribed operator-theoretic procedure* from T_Φ and the boundary condition — a boundary/Wronskian determinant, a regularised Fredholm determinant of the resolvent, or a rigorously convergent half-line limit of truncated transfer determinants — and prove

$$(7) \quad D_{\theta_0}(\lambda) = 0 \iff \lambda \in \sigma_p(T_{\Phi, \theta_0}),$$

with the order of each zero equal to the (separately specified) spectral multiplicity, and with the real normalisation $D_{\theta_0}^\# = D_{\theta_0}$. *PASS condition.* D_{θ_0} may *not* be defined as a canonical product over prescribed points (in particular not as Ξ , nor over $\{\gamma_n\}$); it must arise from T_Φ by the fixed procedure above.

(4) **Divisor identity.** Prove the zero-divisor identity

$$(8) \quad \operatorname{div} D_{\theta_0} = \operatorname{div} \Xi.$$

Given (1)–(4): since the eigenvalues of a self-adjoint realization are real, eq. (8) forces all zeros of Ξ to be real, i.e. RH. The point of the PASS conditions in (1) and (3) is exactly that only then is eq. (8) a genuine *theorem* rather than a definition of the objects in terms of the target.

Remark 3.3 (Direction, and difficulty, of Open Problem 3.2). Only the implication “(1)–(4) $\Rightarrow$ RH” is claimed; the converse is not, and RH does not by itself force any particular T_Φ to have characteristic function Ξ . Thus Open Problem 3.2 is *at least as hard as* RH, and it imposes additional requirements not known to follow from RH: not only that the zeros of Ξ be real but that they coincide, with order, with the eigenvalue divisor of a *specific* operator. Contrary to a claim in an earlier draft of this audit, this does not mean there is no RH-independent acceptance criterion: the identity eq. (8) is perfectly well-posed to *state* without knowing RH, and its target $\operatorname{div} \Xi$ is explicit. What is missing is different — a rigorous definition of T_Φ and D_{θ_0} , and a proof relating the latter to the already explicit target divisor $\operatorname{div} \Xi$ (remark 3.5). This paper resolves none of it.

Remark 3.4 (Three distinct spectral conditions, not one). “spec = $\{\gamma_n\}$ with multiplicities and correct weights”, as an earlier draft put it, conflates three genuinely different requirements that must be separated:

- (a) *support*: $\operatorname{supp} \rho = \{\gamma_n\}$, using the cyclic Weyl spectral measure ρ and its closed support (a statement about the spectrum as a set);
- (b) *spectral multiplicity*: the eigenvalue multiset, i.e. the multiplicity of each point of the spectrum. Under additional hypotheses ensuring simple eigenvalues (which a separated boundary condition on a scalar half-line system typically, but not automatically, provides), matching zeta-zero orders would further force the zeros of Ξ to be *simple* — a statement strictly stronger than RH;
- (c) *norming constants*: the weights $\rho(\{\gamma_n\})$ of the atoms of the spectral measure, which are *not* eigenvalue multiplicities but the squared reciprocal norms of the eigenfunctions, after fixing the normalisation of the canonical eigenfunctions.

None of (a)–(c) is equivalent to another. The support condition (a) alone is insufficient; the divisor identity eq. (8) used here does retain the missing real-part information (other spectral data encoding the real parts could serve equally). No formula for the weights in (c) is available in [12]; that gap is part of remark 3.5.

Remark 3.5 (T_Φ and H_Φ are not yet defined). The roadmap [12] refers to T_Φ “induced by Φ through F1” but supplies no Hamiltonian formula, transfer matrix, Weyl function, or de Branges chain. Consequently one cannot yet speak rigorously of the minimal/maximal relation of T_Φ , its self-adjoint extensions, or its spectral measure: these presuppose H_Φ . The genuinely first task — logically prior to any spectral identification — is therefore the *construction* of H_Φ and T_Φ , stated as obligation (1) of Open Problem 3.2. The target $\operatorname{div} \Xi$ is, by contrast, already explicit; what is missing is a rigorous definition of T_Φ and D_{θ_0} and a proof relating the latter to that target. Until obligation (1) is done, eq. (5) and the left-hand side of eq. (8) refer to objects that are not defined, and the phrases “the Weyl function of T_Φ ” or “the spectral measure ρ ” are placeholders.

Remark 3.6 (Why the log-derivative realization does not solve this). Corollary 2.2 produces a half-line system whose spectral measure is atomic, supported on the zeros of H_t . Even setting aside that it works from H_t ($t > \Lambda$) rather than the putative system constructed from the undeformed kernel Φ , it does not solve Open Problem 3.2: it matches

a *zero set* by construction, whereas Open Problem 3.2 demands the zero-divisor identity eq. (8) for a *specific*, independently constructed T_Φ (and, by the PASS condition of Open Problem 3.2(3), for a D_{θ_0} produced from T_Φ , not defined over prescribed points). Matching a prescribed set of points by choosing a Herglotz function is trivial; proving that a *given* operator has a prescribed characteristic divisor is the whole problem.

Remark 3.7 (Why no terminal entire function is imposed). The D1 model of [12] is a singular canonical system on $[0, \infty)$. The functions E_L associated with finite truncations $[0, L]$ are approximation data for this half-line system; D1 does not assume that E_L converges to a nonzero entire function as $L \rightarrow \infty$. In particular the condition $A_\infty = c\Xi$ is neither required nor used: a nonzero entire terminal function is characteristic of a *finite, regular* chain, whereas a genuine half-line limit-point endpoint typically admits no such terminal entire function. (The characteristic function D_{θ_0} of eq. (8), when it exists, is a different object — attached to the self-adjoint realization, not a terminal transfer-matrix entry.) Imposing $A_\infty = c\Xi$ would convert D1 into a separate finite-chain realization problem, which the programme does not require.

Remark 3.8 (The limit-point conjunct is conditional, not settled). The first conjunct of eq. (5) — limit-point at $+\infty$ — is often described in [12] as the “easy half”, on the ground that the Riemann–von Mangoldt law $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + O(\log T)$ “forces infinite canonical length”. We do *not* endorse that inference. $N(T)$ counts zeros in the *spectral* variable, whereas $\int_0^\infty \text{Tr } H_\Phi dx$ is a length in the *spatial* variable. Such a deduction would require *either* a direct estimate of the total trace from an explicit formula for H_Φ (needing no spectral identification at all), *or* a proved spectral-counting correspondence — weaker than the full divisor identity eq. (8) — combined with a suitable Cartwright/de Branges growth/type theorem converting spectral density into canonical length. (We do not assert that $N(T) \sim \frac{T}{2\pi} \log T$ alone yields trace divergence; a precise such theorem would have to be cited and its hypotheses checked.) Neither is supplied here. Moreover infinite *discrete* spectrum alone does not force infinite length — a finite regular canonical system can have infinite discrete spectrum — so the super-linear $T \log T$ growth would have to be used explicitly to exclude finite exponential type. Accordingly we treat trace divergence as a *separate hypothesis*, not a settled fact:

No theorem translating the Riemann–von Mangoldt counting law into $\int_0^\infty \text{Tr } H_\Phi(x) dx = \infty$ is proved in this paper.

Even granting it, by remark 1.2 it is a reparametrisation invariant that cannot be manufactured by renormalisation, and it must not be re-derived from proposition 4.1 on pain of circularity.

Scope note. OPEN. The following obstacles constrain any resolution of Open Problem 3.2; none is resolved here. They also explain why the heat-flow family H_t ($t > \Lambda$) of section 2 — [12]’s Λ -1/ Λ -2 step, where all zeros are already real — gives no RH-independent handle on the putative system constructed from the undeformed kernel Φ .

Obstacle 1: a real entire function is never Hermite–Biehler. For any real entire F one has $F^\# = F$, so the strict de Branges inequality $|E(z)| > |E^\#(z)|$ ($\text{Im } z > 0$) fails identically. Thus neither Ξ nor H_t can itself be an E -function; the spectral data of the half-line system must be encoded through a genuine Hermite–Biehler $E = A - iB$ (equivalently a Weyl function m), not by declaring a real entire function a structure function.

Obstacle 2: the naive $|H_t|^{-2} d\tau$ is not a spectral measure. For $t > \Lambda$ all zeros of H_t are real, and there are infinitely many [8, 6]. At each real zero $|H_t(\tau)|^{-2}$ has a

non-integrable double pole, so $d\mu_t = |H_t(\tau)|^{-2} d\tau$ is not locally finite; it cannot be the representing measure of any Weyl function.

Obstacle 3: no controlled spectral-system limit toward the candidate parameter $t = 0$. Even a complete identification for H_t at a fixed $t > \Lambda$ would carry no RH information: there the zeros of H_t are already all real ([12] Λ -2), so real-rootedness is not in question. The programme locates RH at the limit down to the candidate parameter $t = 0$ (the true critical parameter is Λ , not known to be 0). Here t is the heat-flow parameter of eq. (1); the roadmap's symbol λ denotes its flow parameter, which agrees with t up to the fixed rescaling recorded after eq. (1). We use t throughout. The roadmap's constraint (C') argues that the relevant minimum should tend to 0 as $t \downarrow 0^+$ with *equality*: a uniform positive margin would, *provided* the requisite continuity of that minimum in the parameter, its strict equivalence with the real-zero property, and a stability/continuation theorem past 0 are established, extend the conclusion below 0 and hence contradict $\Lambda \geq 0$. Those implications are the roadmap's, not results proved here. Note that the function H_t of course has a limit as $t \downarrow 0$; what is missing is any control of the *canonical system or spectral data* along that limit. No such control is provided here.

4. THE LIMIT-POINT CRITERION (REFERENCE STATEMENT)

Proposition 4.1 (Trace-divergence limit-point criterion). *Let $H: [0, \infty) \rightarrow M_2(\mathbb{R})$ satisfy the standing assumptions of [3]: $H \in L^1_{\text{loc}}$, $H \succeq 0$, $H(x) \neq 0$ for a.e. x , and $[0, \infty)$ is not a single indivisible interval. Then the right endpoint $x = +\infty$ of the canonical system eq. (2) is in the limit-point case if and only if*

$$\int_0^\infty \text{Tr } H(x) dx = +\infty.$$

This is a specialisation of [3, Theorem 3.5]; it is stated here only for reference.

Remark 4.2 (On the standing assumptions). The a.e. non-vanishing and no-single-indivisible-interval hypotheses are Remling's standing conventions. They do *not* by themselves turn the maximal/minimal objects into operators: even with $H(x) \neq 0$ a.e., rank-one (indivisible) subintervals force one to work with linear *relations*, which is why we use relation language throughout (cf. [3, Theorem 2.25]). A merely positive-semidefinite H vanishing on a positive-measure set is reduced to this setting by the trace reparametrisation of remark 1.2; by that same invariance the trace integral, and hence the classification, is unchanged.

Proof. By [3, Theorem 3.5], under the standing assumptions the right endpoint is limit-point iff $H \notin L^1((c, \infty); M_2)$ for one — equivalently every — $c > 0$. Since $H \succeq 0$, positivity gives $H_{11}, H_{22} \geq 0$ and $|H_{12}| \leq \sqrt{H_{11}H_{22}} \leq \frac{1}{2} \text{Tr } H$, whence for each $c > 0$

$$H \in L^1((c, \infty); M_2) \iff \text{Tr } H \in L^1(c, \infty).$$

Combining the two equivalences gives the stated trace form. (The reference uses $Jy' = -zHy$; the substitution $z \mapsto -z$ gives the sign convention of eq. (2) and does not affect endpoint classification. The same criterion appears in Winkler [4].)

For the reader's convenience we indicate the direction that has an elementary proof. If $\int_c^\infty \text{Tr } H < \infty$ then $H \in L^1(c, \infty)$; writing eq. (2) as the integral equation $y(x) = y(c) + z \int_c^x J^{-1}H(s)y(s) ds$ and applying Gronwall's inequality, every solution at every fixed z is bounded near $+\infty$, so $\int_c^\infty y(x, z)^* H(x) y(x, z) dx \leq (\sup \|y\|^2) \int_c^\infty \text{Tr } H < \infty$; all solutions are then in L^2_H and $+\infty$ is limit-circle. The reverse implication — trace divergence $\Rightarrow$ limit-point — is *not* elementary in this way: exhibiting a single non- L^2 solution at the special value $z = 0$ (the constant vector spanning the larger of $\int_c^\infty H_{jj}$) does not by itself

control the solution count at non-real z . That step requires the parameter independence of the deficiency index, which is precisely the nontrivial content of [3, Theorem 3.5] via [3, Lemma 3.7]; we do not reproduce it, and rely on the cited theorem for the full equivalence. $\square$

5. CONDITIONAL COROLLARY AND BOUNDARY STRUCTURE

Corollary 5.1 (The limit-point conjunct, conditionally). *Assume the construction of remark 3.5 is carried out, giving a half-line Hamiltonian H_Φ and relation T_Φ satisfying the standing assumptions of proposition 4.1, and assume the separate hypothesis $\int_0^\infty \text{Tr } H_\Phi(x) dx = +\infty$ of remark 3.8. Then $x = +\infty$ is in the limit-point case. This is the first conjunct of eq. (5); it is conditional on both the construction and the trace-divergence hypothesis, the latter to be supplied from the construction and not from proposition 4.1 (remark 3.8).*

Proof. Immediate from proposition 4.1 applied to H_Φ . $\square$

Remark 5.2 (This is not unconditional, and not the trivial case). Corollary 5.1 settles only the first conjunct of eq. (5), and only modulo the construction and the trace-divergence input. The second conjunct is the corrected identification Open Problem 3.2: even in its right form (the divisor identity eq. (8)) it is at least as hard as RH (remark 3.3), and it is *not* addressed. For the trace-normalised realization of corollary 2.2 the trace-divergence hypothesis is automatic but yields only the tautology of remark 2.3. So the paper establishes *nothing* arithmetically meaningful: all content sits in the spectral identification, which we do not resolve.

Corollary 5.3 (Boundary-parameter structure, under corollary 5.1). *Under the hypotheses of corollary 5.1, the minimal symmetric relation T_Φ associated with H_Φ has deficiency indices $(1, 1)$: the right endpoint $x = +\infty$ is limit-point and contributes no free parameter, while the regular left endpoint $x = 0$ carries one self-adjoint boundary condition. For every $\theta_0 \in [0, \pi)$ the condition $e_{\theta_0}^* J y(0) = 0$, with $e_{\theta_0} = (\cos \theta_0, \sin \theta_0)^\top$, defines a self-adjoint realization, and these exhaust all self-adjoint realizations [3, Theorem 2.25]. In particular limit-point at $+\infty$ does not make T_Φ essentially self-adjoint and does not yield a unique self-adjoint realization; this corrects the “ α does not appear / unique self-adjoint realization” phrasing of [12]’s D1, which overstates the consequence of limit-point.*

We phrase this for relations throughout: because indivisible subintervals may occur even when $H_\Phi(x) \neq 0$ a.e. (cf. remark 4.2), the maximal/minimal objects need not be operators. Upgrading to a densely defined operator requires separately showing the multivalued part of the maximal relation is trivial; this is governed by the operator-part / multivalued-part decomposition of the relation (see the corresponding discussion in [3]), not by [3, Theorem 2.25], which concerns only the parametrisation of self-adjoint boundary conditions.

6. DISCUSSION AND SCOPE

6.1. What this paper establishes. The only proved statements are (i) the reference criterion proposition 4.1, a direct specialisation of [3, Theorem 3.5], (ii) the trivial log-derivative realization proposition 2.1–corollary 2.2, and (iii) the *negative* observation that the programme’s Core Lemma eq. (5) is *not justified as written* as an equivalence with RH (remark 3.1). None is a new theorem in analysis: (i) is standard and depends on [3]; (ii) is a routine consequence of the Laguerre–Pólya factorisation; (iii) is an elementary but consequential identification of a logical gap. We therefore present the paper as a reference note plus a logical-gap audit, not a source of new theorems.

We do *not* establish essential self-adjointness of any operator (or relation) attached to the de Bruijn–Newman flow, nor a unique self-adjoint realization, nor any arithmetically meaningful limit-point property of that flow, for any value of t .

6.2. What remains open. The corrected identification problem Open Problem 3.2 is open. By remark 3.3 only the one-way sufficiency “ $\operatorname{div} D_{\theta_0} = \operatorname{div} \Xi \Rightarrow \text{RH}$ ” holds, and it is at least as hard as RH — imposing additional requirements not known to follow from RH, since it also asks that the divisor match that of a *specific* operator (remark 3.4). Logically prior to it is the unstated *construction* of H_Φ and T_Φ (remark 3.5); the target $\operatorname{div} \Xi$ is by contrast already explicit. Until the construction is done, T_Φ , D_{θ_0} , the Weyl function, and the spectral measure — i.e. the left-hand data of eqs. (5) and (8) — refer to objects that are not yet defined.

The limit-point conjunct of eq. (5) is *not* the hard part, but neither is it settled: as recorded in remark 3.8, no theorem translating the Riemann–von Mangoldt law into $\int_0^\infty \operatorname{Tr} H_\Phi dx = \infty$ is proved here, so trace divergence remains a hypothesis — which, if granted, gives limit-point via proposition 4.1, subject to the anti-circularity and invariance cautions of remarks 1.2 and 3.8. We also correct [12]’s D1 write-up on one point of principle: limit-point at $+\infty$ does not give a *unique* self-adjoint realization — the regular left endpoint retains the parameter θ_0 (corollary 5.3).

Finally, even a correct divisor identity eq. (8) presupposes that the realization has discrete spectrum and a characteristic entire function; a general half-line self-adjoint canonical system need have neither (it may have absolutely continuous or singular continuous spectrum), so establishing purely discrete spectrum with no continuous part is itself part of the open problem. None of this is addressed here, and the critical limit (remark 3.8, Obstacle 3 of section 3) lies outside the present scope.

6.3. Aside: the Conrey–Li obstruction and finite-chain realizations. This aside is *not* part of the D1 half-line problem; it concerns the separate finite-chain realization that D1 does not require. The Conrey–Li work [5] concerns specific positivity conditions of de Branges type for ξ/L -functions; it is *not* merely the observation that the real entire candidate $E_T(z) = \xi(1/2 - iz)$ fails the Hermite–Biehler test (that failure is trivial, since $E_T^\# = E_T$). A finite-chain de Branges realization with a terminal A -function proportional to H_t would require building a genuine $E_t = A_t - iB_t$ with nonzero B_t ; this is a separate problem and is *not* needed for D1 (remark 3.7). The precise relation of any such E_t to the Conrey–Li positivity conditions — for which we refer the reader to [5] directly — is beyond this audit. Any earlier claim that $E_\lambda = \Xi_\lambda$ “avoids” the obstruction was mistaken.

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